Predico: an independent study

A symptom-based disease predictor, its data, how well it really works, where it fails, and how it should and should not be used.



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REPOSITORY
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MODEL
Random forest, 300 trees

SCOPE
41 diseases · 131 symptoms

Abstract

Predico is a small web app: a user picks symptoms from a list of 131, and a random forest ranks 41 possible diseases. This report studies the project as it stands in 2026. It audits the data, re-evaluates three models under realistic conditions, analyses errors, checks whether the confidence scores mean anything, and explains what drives the predictions.

The training data contains 4,962 rows but only 305 unique cases. Given a full symptom list, the random forest is almost always right (99.9%), but this reflects a very tidy dataset. When a user reports only 3 symptoms, as real users tend to, the correct disease is the top match **72%** of the time and in the top three **87%** of the time. Errors cluster in diseases with no distinctive symptom: hepatitis D, tuberculosis, hepatitis A and the common cold. The model's confidence scores are strongly *under*-confident: it shows 37% on average while being right 72% of the time. Predico is suitable as a learning and demonstration tool. It is not suitable for diagnosis, triage or any medical decision.

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1 Introduction

Online symptom checkers are among the most common health tools people use. They answer one question: given what I feel, what might this be? Predico is a minimal version of that idea, built with classic machine learning on a public dataset, and kept deliberately small: no account, no API key, no data stored.

This study asks four questions:

- How good is the data, and what does it really represent?
- How accurate is the model when a user gives only a few symptoms?
- Where does it fail, and are its confidence scores trustworthy?
- What is a responsible way to present it?

2 System

The project has two scripts and no server-side state.

- `main.py` loads and cleans the CSVs, cross-validates three models, trains the best on all data and writes `saved_model/model.joblib` and `metrics.json`.
- `app.py` is a Streamlit page. It loads the model once (cached), turns the chosen symptoms into a 131-long yes/no vector, and shows the five most likely diseases with the model's vote share.
- `config.yaml` holds the seed, model settings and evaluation settings.

COMPONENT	VERSION
Streamlit	`1.67.0`
scikit-learn	`1.9.1`
pandas / NumPy	`3.1.1 / 2.5.4`
Hosting	`Streamlit Community Cloud`

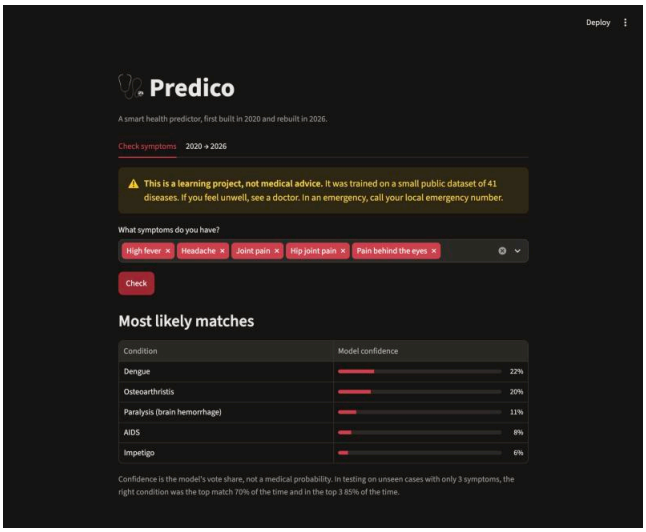


Figure 1. The app ranks five matches with the model's confidence and a medical disclaimer.

3 Data audit

The data is the widely shared "Disease Prediction Using Machine Learning" set from Kaggle [1]: rows of yes/no symptoms, each labelled with one disease.

MEASURE	VALUE
Rows in the CSVs (training + test)	4,962
Unique symptom patterns	305
Diseases	41
Unique cases per disease	5 – 10
Symptoms per case (min / median / max)	3 / 7 / 17
Symptoms that occur in only one disease	78 of 131
Most shared symptom	vomiting, 17 diseases

Findings

- **94% of rows are duplicates.** The original test file had 42 rows, and 41 of them also appear in training. Any score measured on that split is meaningless, a common form of leakage [9]. All analysis here uses the 305 unique cases.
- **A broken column.** `fluid_overload` is always zero; the real values sit in a duplicate column, `fluid_overload.1`. Cleaning merges them, leaving 131 symptoms.
- **It looks synthetic.** 78 symptoms belong to exactly one disease, and many diseases have a symptom present in 100% of their cases (every allergy case has shivering). Real patient records are never this clean, so scores here are an upper bound.
- **Sensitive and odd features.** "Extra marital contacts" is a feature for AIDS, which encodes stigma rather than clinical risk. Labels contain typos ("Osteoarthristis", "Peptic ulcer diseae", "hemmorhoids"), and capitalisation is inconsistent ("hepatitis A").
- **No context.** There is no age, sex, duration, severity or history, all of which doctors rely on.

4 Method

- **Models:** decision tree (entropy), random forest (300 trees) [3] and Bernoulli naive Bayes ($\alpha = 1$), all from scikit-learn [2].
- **Validation:** stratified 5-fold cross-validation on the 305 unique cases, repeated 3 times with different seeds (915 predictions per setting).
- **Realistic input:** for each held-out case, keep only k of its symptoms, chosen at random, for $k = 1$ to 6. This simulates a user who reports what bothers them most and stops.
- **Metrics:** top match (top-1 accuracy), in top 3 (top-3 accuracy), and expected calibration error (ECE) [4].

The app's own "70% / 85%" figure comes from `main.py`, which runs one repeat. This report averages three, giving 72% / 87%. The difference is sampling noise.

5 Results

MODEL	ALL SYMPTOMS	3 SYMPTOMS, TOP MATCH	3 SYMPTOMS, IN TOP 3
Decision tree	46.4%	15.4%	21.1%
Random forest	99.9%	72.1%	87.2%
Naive Bayes	97.9%	61.9%	74.1%

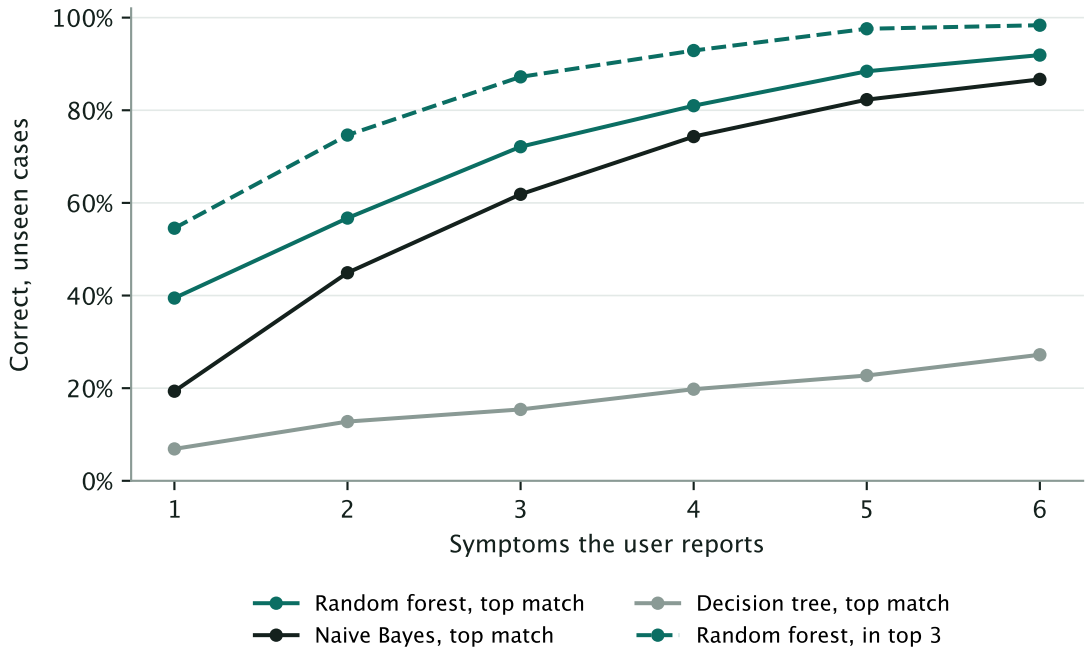


Figure 2. Accuracy on unseen cases as the user reports more symptoms. Each point is 915 predictions.

> **The number of symptoms matters more than the model.** With one symptom, the random forest is right 40% of the time. Each extra symptom adds a lot until about five, where it passes 88%. That is why the app warns users who enter fewer than three.

A single decision tree fails because each case has only 5–10 siblings: one tree memorises exact patterns and cannot cope with missing symptoms. A forest of 300 trees, each built on different symptoms, degrades gracefully. Naive Bayes sits in between, which fits its assumption that symptoms are independent.

6 Error analysis

With 3 symptoms, the random forest gets 22 of 41 diseases right at least 95% of the time. The rest fail in a clear pattern.

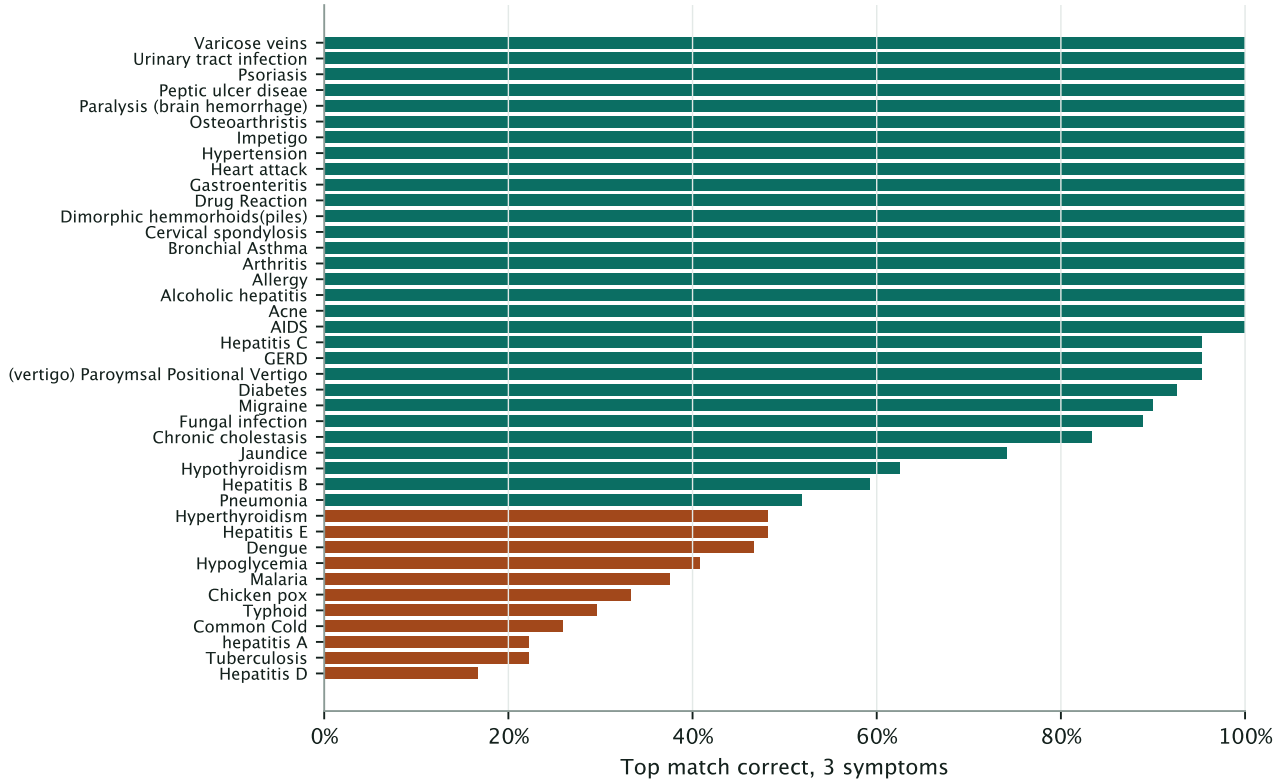


Figure 3. Top-match accuracy per disease with 3 symptoms (random forest). Orange: below 50%.

ACTUAL	PREDICTED AS	RATE
Hepatitis D	Hepatitis C	40%
Tuberculosis	Heart attack	33%
Chicken pox	Fungal infection	30%
Pneumonia	Heart attack	30%
Hepatitis A	Gastroenteritis	26%
Malaria	Gastroenteritis	25%
Common cold	Allergy	22%
Hyperthyroidism	Heart attack	22%

> **Safety note.** Mixing up tuberculosis and heart attack is not harmless in either direction. This is why the app shows five ranked options and never a single verdict.

Why these fail

- **No giveaway symptom.** The worst diseases (hepatitis D, hepatitis A, malaria, chicken pox) have no symptom unique to them; everything they show is shared with others (Appendix A).
- **Hepatitis family.** Hepatitis A, C, D and jaundice share yellow skin, dark urine and nausea. With 3 symptoms they are hard to tell apart, for a doctor too.
- **"Heart attack" as a catch-all.** Chest pain, breathlessness and sweating appear across lung and thyroid diseases, so the model often lands on heart attack.

7 Calibration

The app shows a confidence next to each match. A well-calibrated model that says "40%" should be right about 40% of the time. Figure 4 plots stated confidence against how often the top match was actually right, with 3 symptoms.

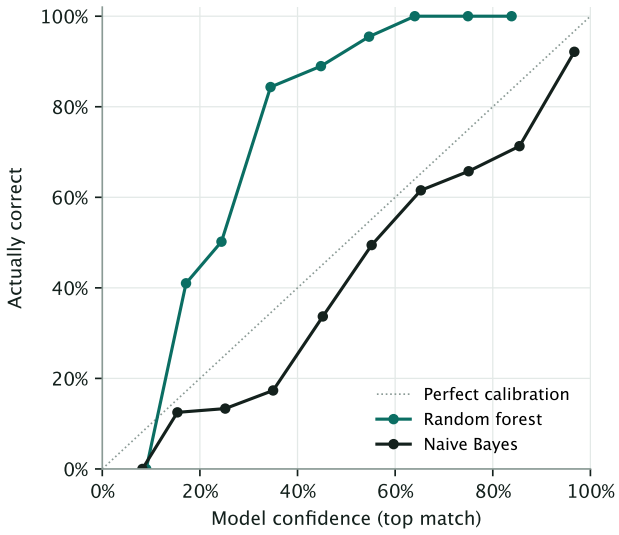


Figure 4. Reliability diagram, 3 symptoms. Above the dotted line means under-confident.

	RANDOM FOREST	NAIVE BAYES
Average confidence	37%	70%
Actually right	72%	62%
Calibration error (ECE)	0.35	0.08
Shown below 30%	42%	8%

> **The random forest undersells itself.** When it shows 30–40%, it is right 84% of the time. Its "confidence" is the share of trees that voted for that disease, and with 41 options those votes spread thinly.

Naive Bayes is less accurate but much better calibrated. For users, under-confidence is the safer failure, but the numbers are still misleading. The app labels them "model confidence, not a medical probability", which is accurate. Section 13 suggests a fix.

8 Explainability

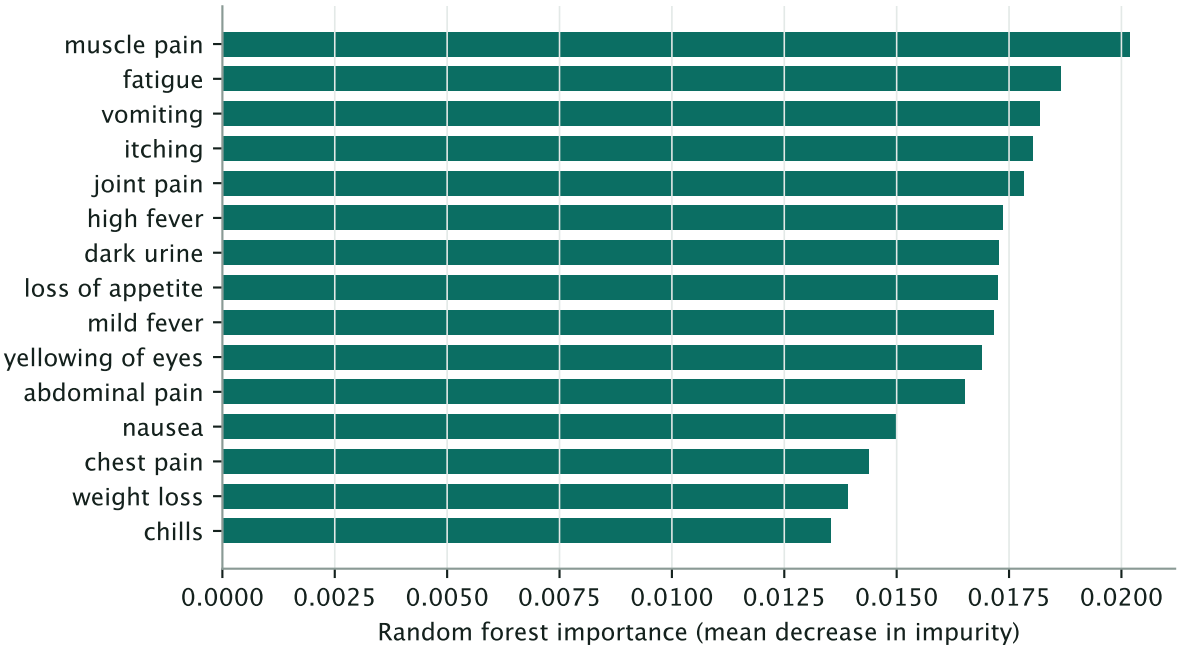


Figure 5. The 15 symptoms the random forest relies on most overall.

The most important symptoms overall are broad ones (muscle pain, fatigue, vomiting, itching, joint pain). They do not identify a disease alone but split the 41 into groups. The deciding step is usually a disease's *signature symptom*: one that, when present, points almost only to that disease (for example, "silver-like dusting" for psoriasis or "rusty sputum" for pneumonia). Appendix A lists these for every disease. Diseases without one are mostly the ones that fail in the error analysis.

9 Model card

Following the model card format of Mitchell et al. [5].

Model	Random forest classifier, 300 trees, scikit-learn 1.9.1, seed 42. Trained on 305 unique cases.
Input	Yes/no presence of up to 131 symptoms.
Output	Vote share across 41 diseases; the app shows the top 5.
Intended use	Learning and demonstration: showing how symptom-based classification works and where it breaks.
Out of scope	Diagnosis, triage, deciding whether to seek care, any clinical or insurance decision, use on children or in pregnancy, any disease outside the 41.
Performance	3 symptoms: 72% top match, 87% in top 3. 1 symptom: 40% / 55%. Full symptom list: 99.9% (inflated by tidy data).
Known weaknesses	Hepatitis D and A, tuberculosis, common cold, typhoid, chicken pox and malaria below 40% with 3 symptoms. Confidence understated by about half.
Data caveats	Likely synthetic; no demographics, duration or severity; one stigmatising feature; typos in labels.
Fairness	Cannot be assessed: the data has no demographic fields.
Privacy	The app stores nothing; symptoms stay in the browser session.

10 Responsible use and regulation

- **Not a medical device, by design.** In the EU and US, software intended to diagnose or guide treatment can be a regulated medical device. Predico is explicitly not intended for that, and says so on the page.
- **EU AI Act.** The Act (Regulation (EU) 2024/1689) [6] treats AI that is a medical device, or part of one, as high-risk, with duties for data quality, documentation, human oversight and accuracy. Predico stays outside that scope only because of its stated purpose. A health product built on it would not.
- **FDA.** The FDA now publishes guidance and a public list of AI-enabled medical devices [7]. Symptom checkers that only give general information are usually not regulated as devices, but the line depends on intended use and claims.
- **Evidence from the field.** An early audit of 23 commercial symptom checkers found the right diagnosis listed first only about a third of the time [8]. Predico's tidy-data numbers should not be compared to that directly.
- **Design choices that follow.** Clear disclaimer, ranked options instead of a verdict, a warning for low input, no data stored, and an emergency message.

11 Limitations

- All results come from a small, likely synthetic dataset. Real-world accuracy would be lower.
- The symptom-dropping test picks symptoms at random. Real users report the most bothersome ones first, which may be more or less informative.
- Only 41 diseases. Anything else is forced into one of them; the model cannot say "none of these".
- Symptoms are yes/no only, with no severity, duration or timing.
- Feature importance from impurity can favour common symptoms; it describes the model, not medicine.

12 Reproducibility

Every number in this report can be regenerated from the repository. Seeds are fixed at 42 (plus 43 and 44 for the extra repeats).

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt matplotlib
python main.py                      # trains, writes saved_model/
python docs/report/analysis.py      # figures + docs/report/results.json
# print docs/report/report.html to PDF, e.g.:
chrome --headless --print-to-pdf=docs/Predico-2026-report.pdf docs/report/report.html
streamlit run app.py
```

Raw results: `docs/report/results.json`. Versions are pinned in `requirements.txt`.

13 Future work

- **Fix the confidence numbers.** Calibrate the forest (for example with scikit-learn's `CalibratedClassifierCV`), or show naive Bayes probabilities alongside.
- **Allow "none of these".** Flag inputs that don't resemble any training case instead of forcing a match.
- **Better data.** A dataset with real clinical cases, demographics and severity, such as published vignette sets used to evaluate symptom checkers [8].
- **Ask the next question.** Suggest the one symptom that would best separate the top matches, the way a doctor would ask.
- **Natural-language input.** Let users describe symptoms in their own words and map them to the 131 features, keeping the transparent model underneath.

A Signature symptoms

For each disease: its common symptoms (present in at least half its cases), ranked by how strongly each one points to that disease alone. "100%" means every case with that symptom has this disease.

DISEASE	MOST TELLING SYMPTOMS	3-SYMPTOM ACCURACY
(vertigo) Paroymsal Positional Vertigo	spinning movements 100%, unsteadiness 100%, loss of balance 38%	95%
Acne	pus filled pimples 100%, blackheads 100%, scurring 100%	100%
AIDS	muscle wasting 100%, patches in throat 100%, extra marital contacts 100%	100%
Alcoholic hepatitis	swelling of stomach 100%, distention of abdomen 100%, history of alcohol consumption 100%	100%
Allergy	shivering 100%, watering from eyes 100%, continuous sneezing 33%	100%
Arthritis	movement stiffness 100%, swelling joints 45%, painful walking 45%	100%
Bronchial Asthma	mucoid sputum 100%, family history 46%, breathlessness 23%	100%
Cervical spondylosis	weakness in limbs 100%, neck pain 45%, back pain 33%	100%
Chicken pox	red spots over body 50%, swelled lymph nodes 35%, mild fever 33%	33%
Chronic cholestasis	itching 17%, yellowing of eyes 12%, yellowish skin 11%	83%
Common Cold	throat irritation 100%, loss of smell 100%, congestion 100%	26%
Dengue	pain behind the eyes 100%, back pain 67%, red spots over body 45%	47%
Diabetes	irregular sugar level 100%, increased appetite 100%, polyuria 100%	93%
Dimorphic hemmorhoids(piles)	pain during bowel movements 100%, pain in anal region 100%, bloody stool 100%	100%
Drug Reaction	spotting urination 100%, burning micturition 56%, stomach pain 45%	100%
Fungal infection	nodal skin eruptions 100%, dischromic patches 100%, itching 12%	89%
Gastroenteritis	sunken eyes 100%, dehydration 100%, diarrhoea 11%	100%
GERD	ulcers on tongue 100%, stomach pain 55%, acidity 40%	95%
Heart attack	breathlessness 15%, chest pain 10%, sweating 9%	100%
hepatitis A	mild fever 33%, muscle pain 26%, diarrhoea 23%	22%

DISEASE	MOST TELLING SYMPTOMS	3-SYMPTOM ACCURACY
Hepatitis B	yellow urine 100%, receiving blood transfusion 100%, receiving unsterile injections 100%	59%
Hepatitis C	family history 46%, yellowing of eyes 10%, yellowish skin 10%	95%
Hepatitis D	dark urine 22%, joint pain 20%, yellowing of eyes 16%	17%
Hepatitis E	acute liver failure 100%, coma 100%, stomach bleeding 100%	48%
Hypertension	lack of concentration 100%, loss of balance 31%, dizziness 29%	100%
Hyperthyroidism	muscle weakness 64%, mood swings 53%, abnormal menstruation 53%	48%
Hypoglycemia	anxiety 100%, drying and tingling lips 100%, slurred speech 100%	41%
Hypothyroidism	weight gain 100%, cold hands and feets 100%, puffy face and eyes 100%	63%
Impetigo	blister 100%, yellow crust ooze 100%, red sore around nose 83%	100%
Jaundice	weight loss 25%, dark urine 20%, itching 19%	74%
Malaria	muscle pain 23%, diarrhoea 20%, sweating 16%	38%
Migraine	visual disturbances 100%, stiff neck 64%, acidity 60%	90%
Osteoarthristis	knee pain 100%, hip joint pain 100%, neck pain 55%	100%
Paralysis (brain hemorrhage)	weakness of one body side 100%, altered sensorium 100%, headache 5%	100%
Peptic ulcer diseae	passage of gases 100%, internal itching 100%, indigestion 40%	100%
Pneumonia	rusty sputum 100%, fast heart rate 53%, breathlessness 31%	52%
Psoriasis	silver like dusting 100%, small dents in nails 100%, inflammatory nails 100%	100%
Tuberculosis	blood in sputum 100%, swelled lymph nodes 35%, phlegm 35%	22%
Typhoid	toxic look (typhos) 100%, belly pain 100%, constipation 62%	30%
Urinary tract infection	bladder discomfort 100%, foul smell of urine 100%, continuous feel of urine 100%	100%
Varicose veins	cramps 100%, bruising 100%, swollen legs 100%	100%

Highlighted rows are below 50% top-match accuracy. Most of them have no symptom that points to them alone. The others (such as hepatitis E) do, but lose it when only 3 symptoms are kept.

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